

9.1 PHOTOCHEMICAL REACTIONS

of fragments, which recombine to give a variety of products. In such nonspecific situations the stoichiometry is by no means straightforward, and the term radiolysis is more appropriate.

9.1 PHOTOCHEMICAL REACTIONS

That chemical reaction can be induced by electromagnetic radiation has been known since the 18th century. A number of important investigations, carried out over a period of many years, led to our present understanding of the nature of photochemical processes. One of these was that of von Grotthuss,¹ who in 1819 showed that electromagnetic radiation to bring about chemical change it must first be absorbed. This conclusion was rediscovered in 1841 by Draper² and often is referred to as the *Grotthuss-Draper law*.

Important quantitative studies on the photochemical reaction between hydrogen and chlorine were made by Dalton³ and later by Wittwer.⁴ These and many subsequent investigations, such as those of Bunsen and Roscoe⁵ on the same reaction, showed that the rate is proportional to the intensity of the light absorbed by the system [compare Eq. (8.63)]. With other reactions the dependence on the intensity of light absorbed is different. With the hydrogen-bromine reaction, for example, the rate is proportional to the square root of the intensity of light absorbed [Eq. (8.57)]. In all cases it is the light absorbed, and not the incident intensity, that directly controls the rate.

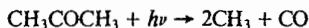
With the advent of the quantum theory and particularly after Einstein's⁶ realization that light can be regarded as a beam of particles (later to be called photons), it became possible to relate the number of reactant molecules transformed in a photochemical reaction to the number of photons absorbed. Stark⁷ in 1911 and independently Einstein⁸ in 1912 proposed that one photon of radiation is absorbed by one molecule. This conclusion, usually referred to as *Einstein's law of photochemical equivalence*, is discussed later in more detail.

9.1.1 Photochemical Primary Process

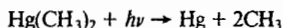
The process in which electromagnetic radiation is absorbed by a molecule, with the formation of species that undergo further reaction, is known as the photochemical primary process. Absorption of radiation leads initially to the formation of an excited species, which may in a subsequent step produce atoms and free radicals; the ways in which this can occur are discussed later. In photochemistry it is usual to consider the formation of the excited species and the subsequent formation of atoms and free radicals as together constituting the primary process. The production of atoms and free radicals in the primary process is referred to as *photolysis* (Greek *photo*, meaning light, and *lysis*, meaning splitting). These atoms and free radicals may be involved in subsequent reactions, which may involve chains; such reactions are referred to as secondary processes. The term photolysis is applied also to an overall photochemical process in which a larger molecule is broken down into smaller ones.

Reactions in which alkyl radicals are produced from organic molecules by the action of ultraviolet radiation are examples of photochemical primary processes. Ace-

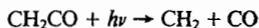
tone, for example, absorbs radiation of wavelengths in the neighborhood of 280 nm to give CH_3CO and CH_3 , and at higher temperatures the acetyl radical further splits into CH_3 and CO . The overall primary process is then



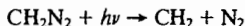
where $h\nu$ represents a photon. The photolysis of the mercury alkyls also has been used as a source of free radicals:



Methylene is produced conveniently by the photolysis of ketene,



and of diazomethane,



A number of important principles relate to the absorption of electromagnetic radiation with the formation of excited molecules and of atoms and free radicals. In the first place, as already noted, the light must be absorbed in order to have any effect; however, not all the absorbed radiation leads to chemical change.

The probabilities of transitions when light is absorbed are governed by the *Franck-Condon principle*. This was first expressed qualitatively by Franck,⁹ who based his conclusions on the relative speeds of electronic transitions and of vibrational motions. The absorption of radiation may occur in 10^{-15} – 10^{-18} s and is so rapid compared to vibrational motion ($\approx 10^{-13}$ s) that immediately after an electronic transition has occurred the internuclear distances in a molecule are much the same as before. Figure 9.1 shows a number of possible arrangements of potential-energy curves for a diatomic molecule, and Franck argued that the most probable transitions are those that can be represented by more or less vertical lines in the diagrams. Radiation of frequency ν can be absorbed by a molecule only if the energy $h\nu$ is equal to the energy difference between a quantized energy level in the excited state and a quantized level in the ground state. The probability of a transition is small if the two states correspond to two very different interatomic distances, since a considerable change in the molecule would have to accompany absorption. The probability also is small if certain selection rules are not satisfied; this is the case, for example, if the two states correspond to different spectroscopic multiplicities, for example, if a singlet $\rightarrow$ triplet transition is involved.

Franck's ideas were put on a more rigorous quantum-mechanical basis by Condon.¹⁰ An important matter relates to the most probable internuclear distances in vibrating molecules. According to classical mechanics, the distance is changing more slowly near the turning points of the motion: for example, a vibrating diatomic molecule spends more time in its fully extended or fully compressed configurations than in intermediate configurations. In quantum mechanics, on the other hand, the situation is different. For the lowest (zero-point) vibrational state the probability is a maximum at the center of the vibration. In the diagrams in Fig. 9.1 the connecting arrows therefore start at the centers of the lines representing the zero-point levels for the ground electronic states. For other vibrational states the maximum is slightly inside each end of the classical vibrational amplitude, and for such states the arrows in Fig. 9.1 end close to

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The characteristic feature of a predissociation spectrum is that there is a diffuse region, corresponding to a transition to the repulsive state followed by rapid dissociation, and that at both longer and shorter wavelengths the spectrum shows a rotational fine structure. This is a different situation from that found with ordinary dissociation (e.g., Fig. 9.1c), where the rotational fine structure can appear only at the lower-energy (longer-wavelength) side of the diffuse bands.

Figures 9.1 and 9.2 relate to a diatomic molecule AB , but the principles are somewhat the same for more complex molecules. The curves in Figs. 9.1 and 9.2 may be regarded as sections through the multidimensional potential-energy curves for polyatomic molecules. However, some additional factors must be taken into account for molecules having more than two atoms. If a diatomic molecule has enough energy to undergo dissociation, it must do so within the period of its first vibration (i.e., within 10^{-13} – 10^{-14} s). On the other hand, as discussed with reference to unimolecular reactions (Section 5.3), a polyatomic molecule having sufficient energy may not dissociate in the first vibration. This is because the energy may not be distributed suitably among the normal modes for bond rupture to occur. As the molecule vibrates, the energy is redistributed constantly among the normal modes, and eventually a vibration leads to dissociation. As a result, a polyatomic molecule with sufficient energy to dissociate may survive for 10^{-8} s or more, and during that time a number of processes may occur, including the emission of radiation.

9.1.2 Reactions of Electronically Excited Species

After an electronically excited molecule has been produced by the absorption of radiation, it can undergo a number of processes, some of which have already been mentioned. The most important of these processes are:

1. *Fluorescence*. The excited molecule may emit radiation and pass into a lower state which may be the ground state or a lower excited state. This process is referred to as *fluorescence*. Usually, this term is restricted to processes in which the molecule does not change its multiplicity, in contrast to *phosphorescence* where there is a multiplicity change (e.g., triplet $\rightarrow$ singlet) and in which the emission is slower.
2. *Molecular dissociation*. This has been considered already with reference to Figs. 9.1 and 9.2.

In the case of fairly simple gas molecules, these two processes are generally the only ones that can occur. With more complex molecules, however, the dissociative processes are much slower, for reasons just discussed. In the liquid phase, collisions may occur within 10^{-12} – 10^{-14} s, which is much shorter than the time required for fluorescence or molecular dissociation to take place. Other processes therefore may be important, such as the following:

3. *Vibrational relaxation*. During this process the molecule passes into lower vibrational states within the same electronic state.

4. *Collisional quenching or external conversion.* Upon collision with other molecules, such as solvent molecules, the electronically excited molecule may pass into a lower electronic state, which is often the ground state. The electronic energy is converted in this process into translational and internal energy (vibrational and rotational) and eventually is dissipated as heat.

In addition, a complex molecule in an electronically excited state may undergo *internal quenching* or *internal conversion*. These processes do not involve collisions; the electronic energy is converted into vibrational energy of the molecule itself, which therefore acts as a self-quencher.

Some of the processes that may occur with an electronically excited molecule are summarized in Fig. 9.3, which is known as a *Jablonski diagram*. All these possible processes have to be considered in a detailed photochemical study. Often the processes leading to atoms and free radicals are regarded together as constituting the primary photochemical act, and subsequent processes are referred to as secondary processes. However, it is to be noted that with this definition the primary act itself may be composite. It is particularly important to appreciate that the excited species formed by the absorption of radiation may undergo processes other than the formation of

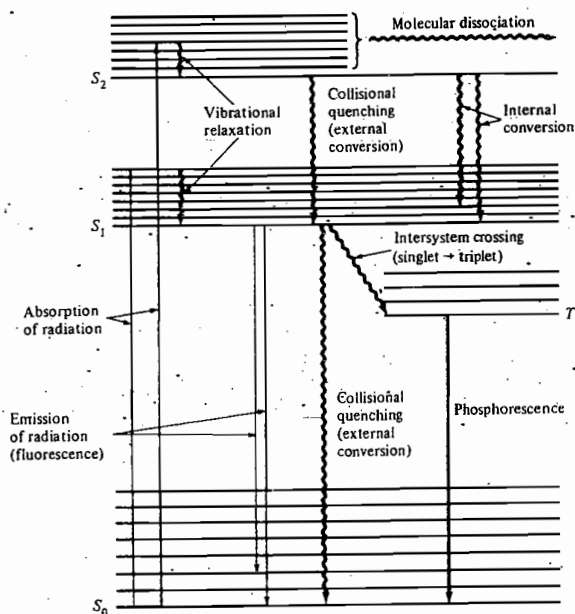


Figure 9.3 Jablonski diagram, which shows the ground state S_0 , a singlet state, and also two excited singlet states (S_1 and S_2) and a triplet state T . Various possible processes are indicated: solid lines are radiative, wavy lines are radiationless.

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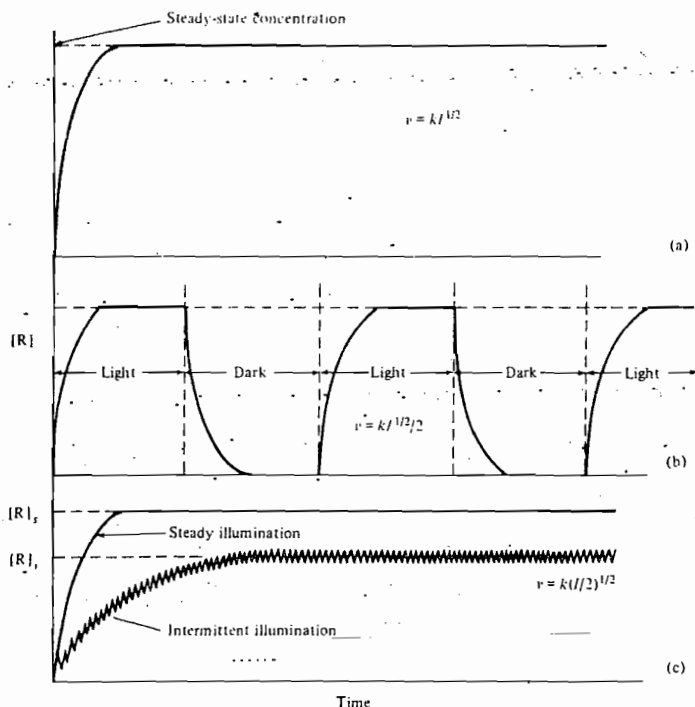


Figure 9.4 Radical concentrations as a function of time: (a) constant illumination; (b) intermittent illumination with equal periods of light and darkness that are long compared with the half-life of the radicals; and (c) intermittent illumination with light and dark periods that are much shorter than the half-life.

Suppose now that the slotted disk is constructed in such a way that the periods of light and darkness are equal. Consider first the case in which the rotation is so slow that the periods of light and dark are much longer than the half-life of the radicals. The variation of radical concentration is now as shown in Fig. 9.4b. During the period of illumination the radical concentration rapidly attains its steady-state value, and during the dark period it falls to zero. Thus, the steady-state concentration of radicals is maintained for only half the period of the experiment, and the rate therefore is one-half of the value for constant illumination. Thus, if the rate with constant illumination is $kI^{1/2}$, that with slow intermittent illumination, with equal light and dark periods, is $kI^{1/2}/2$.

The situation, however, is different if the rate of rotation is so rapid that the periods of light and dark (again assumed equal) are much smaller than the radical half-life. The concentration of radicals now varies as shown in Fig. 9.4c. During the periods of darkness the concentrations decay only to a small extent, and eventually a steady state is reached at which the radical concentration is less than that which cor-

(9.1)

thus

(9.2)

responds to steady illumination with light of the same intensity. In the limit of infinitely rapid rotation, the effect is the same as if the light intensity were $I/2$ and the rate of reaction in this case is $k(I/2)^{1/2}$. This is greater by a factor of $2^{1/2} = 1.41$ than for the case of low rotation speeds.

Therefore, as the rotation speed is varied, the overall reaction rate changes. The transition from the higher rates with rapid rotation to lower rates with slower rotation varies over a range of speeds that are related to the half-life of the radicals. In a photochemical experiment, the rate of formation of the radicals is known (from the rate of absorption of photons), and this rate is equal to the rate of the combination reaction. From this rate and the half-life of the radicals, the concentration of radicals and the rate constant can be calculated.

Shepp's¹⁸ theoretical treatment led to the result that if the ratio of darkness to light is r , the rate for slow rotation is the fraction

$$\frac{1}{1+r}$$

of the rate for steady illumination. For rapid rotation the ratio is

$$\frac{1}{(1+r)^{1/2}}$$

For equal intervals $r = 1$ and the ratios are $\frac{1}{2}$ and $(\frac{1}{2})^{1/2}$, as already seen. If the sector design is such that the period of darkness is three times as long as the period of illumination ($r = 3$), the ratios are $\frac{1}{4}$ and $\frac{1}{2}$, respectively; the rate with rapid rotation is twice that with slow rotation. This case is illustrated in Fig. 9.5, which is based on Shepp's analysis. Curves such as this are displaced to one side or the other if the half-

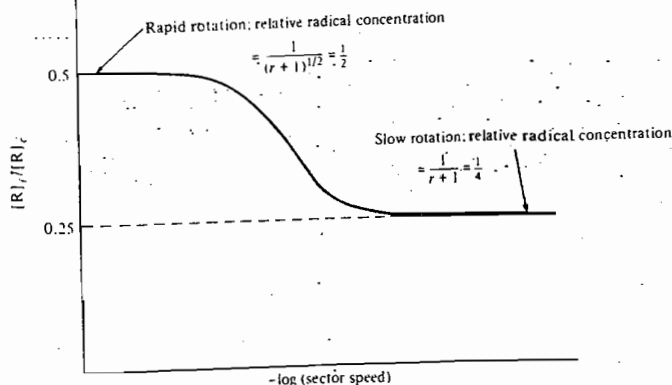


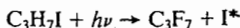
Figure 9.5 Variation of $[R]/[R]_c$, the ratio of the steady-state concentrations for intermittent and constant illuminations, with sector speed. The ratio r of dark to light intervals is 3.

There are three special features of laser radiation. In the first place, it is highly monochromatic, covering only a very narrow range of wavelengths. Also, with a suitable geometry for the laser medium (e.g., the ruby), the beam can be extremely narrow and is said to be *coherent*; the beam divergence can be of the order of a milliradian. Finally, for reasons already explained, the highly coherent and monochromatic beam can be of very high intensity.

These special features of laser beams are of great value in both spectroscopy and photochemistry. The fact that laser beams are highly monochromatic but are nevertheless coherent and of high energy means that atoms and molecules can be excited selectively to an extent that was not possible with ordinary radiation. Also, with laser radiation an atom or molecule can become excited in stages, by the stepwise absorption of a number of photons.

9.2.1 Pulsed Lasers

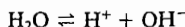
In pulsed lasers the radiation is emitted for periods of very short duration. An important and useful example of a pulsed laser is the *atomic iodine laser*, first used by Kasriel and Pimentel.²¹ This is an example of a *chemical laser*, the effective species being iodine atoms produced photolytically. They are often produced in an excited state by the irradiation of heptafluoropropyl iodide, C_3F_7I , with ultraviolet radiation:



The transition between these excited iodine atoms and ground-state iodine atoms corresponds to a wavelength of 1315 nm, which is in the infrared. Irradiation of the system with such radiation therefore induces laser action, with the production of a powerful coherent beam of 1315 nm wavelength. The photolysis of the C_3F_7I can lead to almost complete dissociation into $C_3F_7 + I^*$. It is possible to produce a large population inversion with this system and hence to obtain a laser beam of very high intensity.

Various techniques have been used to shorten the duration of the pulse emitted by the iodine laser.²² In the "free-running" mode the duration is from 2 to 6 μ s. There is also a "Q-switched" mode, involving a fast shutter, which reduces the duration to 13–30 ns (1 ns = 10^{-9} s), while a "mode-locked oscillator" system produces pulses of 0.2–2 ns duration.

Pulses of such short durations are important in allowing the fastest of chemical reactions to be studied. One application of the iodine laser is an investigation,²³ using the temperature-jump technique, of the rate of the dissociation of H_2O into H^+ and OH^- ions, and the reverse recombination:



The temperature jump was brought about by the absorption of the iodine laser radiation, which rapidly produces vibrations and rotations in the water molecules. The jump occurs as fast as the emission time of the laser, so that the temperature change has been brought about in as short a time as 0.2 ns (2×10^{-10} s). The reaction was followed by observing the conductivity of the solution, the work being done over a range of temperatures. Some of the results of this investigation were referred to in Section 6.7.3.

Pulsed lasers have now been developed in which the duration of the pulse is less than a picosecond (10^{-12} s). Such instruments have been used to study molecular relaxation processes.²⁴

2.2 Multiphoton Excitation

In conventional photochemical experiments there is only a low probability that a molecule excited by the absorption of one photon can absorb an additional one; this is because of the very short lifetimes of excited molecules and because of the relatively small number of photons in an ordinary beam of electromagnetic radiation. The unlikelihood of multiple photon absorption is, in fact, the basis of the law of photochemical equivalence.

With lasers, on the other hand, the situation is very different, because of the much higher intensities of laser beams. There is now a substantial probability that a molecule may absorb successively a number of photons and pass in a stepwise fashion to higher excited states. In fact, in a laser beam even infrared radiation, in which each photon carries only small amounts of energy and can bring about only vibrational and rotational excitation, is capable of causing molecular dissociation as a result of the absorption of a number of photons. With infrared radiation of wavelengths suitable for bringing about vibrational excitation, 20–40 photons are usually sufficient to bring about the breaking of a chemical bond.

The process of infrared multiphoton excitation is illustrated schematically in Fig. 9.7. Three energy regions are to be distinguished. For the first few vibrational states (typically from $v = 0$ to $v = 6$) the vibrational levels are discrete, but for higher levels the states overlap owing to anharmonicity and form the so-called "quasicontinuum." Beyond the level for dissociation the states form a true continuum. Once the molecule has absorbed photons that take it into the quasicontinuum, further excitation takes place more readily because of the availability of suitable energy spacings. Then the system can easily pass into the region of the true continuum, where dissociation can occur.

Much work is being done in this field, and it has led to important contributions to the understanding of unimolecular reactions (see also Section 5.3.9). For an account of some of the kinetic studies on infrared multiple photon excitation a review by Ashfold and Hancock²⁵ should be consulted.

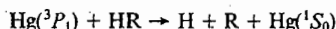
9.3 PHOTOSENSITIZATION

The dissociation of molecular hydrogen into two hydrogen atoms requires energy of 432 kJ mol^{-1} , which corresponds to a wavelength of 277.6 nm. However, as seen from the potential-energy curves in Fig. 9.8, hydrogen does not absorb at all in this region of the ultraviolet, but only at much shorter wavelengths corresponding to energies of more than 1000 kJ mol^{-1} . Working at such short wavelengths is inconvenient, so that alternative methods are needed. A useful technique is *photosensitization*, in which the radiation is absorbed by another species, which transfers energy to a hydrogen molecule on collision.

TABLE 9.1 AMOUNTS OF ENERGY LIBERATED BY EXCITED ATOMS

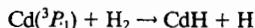
Metal	Transition	Wavelength/nm	Energy of excited atoms/kJ mol ⁻¹	Energy liberated if hydride is formed/kJ mol ⁻¹
Mercury	6 ³ P ₁ -6 ¹ S ₀	253.7	469.4	505.0
	6 ¹ P ₁ -6 ¹ S ₀	184.9	643.9	679.5
Cadmium	5 ³ P ₁ -5 ¹ S ₀	326.1	365.3	430.1
	5 ¹ P ₁ -5 ¹ S ₀	228.8	520.5	585.3
Zinc	4 ³ P ₁ -4 ¹ S ₀	307.6	387.0	483.7
	4 ¹ P ₁ -4 ¹ S ₀	213.9	558.1	654.8

Excited mercury also has been caused to react directly with other substances, such as hydrocarbons. With the saturated hydrocarbons the main primary process that takes place is



where R is a free radical, such as ethyl in the case of ethane. With unsaturated hydrocarbons the same type of process may occur, but in addition an excited molecule can be formed and can undergo subsequent reactions.

Work also has been done using mercury in its singlet-excited 6 ¹P₁ state, which corresponds to a wavelength of 184.9 nm and an energy of 643.9 kJ mol⁻¹. Zinc and cadmium, in both singlet- and triplet-excited states, have been used in photosensitization experiments. The amounts of energy liberated in each case are given in Table 9.1, the values with and without the formation of the hydride being included. By the use of these and other excited states the energy available for transfer to the reacting molecule can be varied over a convenient range. Sometimes there is ambiguity as to whether the hydride is formed or not, although it appears that when the reacting molecule requires the additional energy, the hydride will be formed in order to make it available. This may be illustrated by comparing the reactions of triplet mercury and cadmium. Triplet mercury has sufficient energy to split the H-H bond without hydride formation and, as has been seen, the hydride is not formed as a final product. With triplet cadmium, on the other hand, the energy available is only just about sufficient even if the hydride is formed. The fact that the reaction proceeds readily therefore requires that the hydride be a final product, and this has been found spectroscopically to be the case by Bender.³¹ The overall process is therefore



and other reactions proceed in the same way. For example, Steacie and LeRoy³² detected CdH in the cadmium-photosensitized reactions of propane.

9.4 RADIATION-CHEMICAL REACTIONS

The field of radiation chemistry involves the study of chemical reactions brought about by radiation in which the photons or other particles are of higher energy than

in photochemistry. As previously noted, such radiations bring about some ionization in reaction systems and are less specific in their action than is the case with photochemical processes. The study of radiation-chemical reactions began in about 1935, partly as a result of the construction of particle-accelerating machines at about that time. An additional reason for the interest in the processes was the realization³³ that atoms and free radicals are important products of the primary radiation-chemical step and that these frequently are involved in chain processes. Thus, the field of radiation chemistry is related closely to that of photochemistry and to the study of thermal reactions.

A matter of importance in an experimental study of a reaction induced by ionizing radiations is the rate of absorption of energy by the reaction system: this quantity is commonly expressed in electronvolts (eV) per dm³ per second (1 eV corresponds to 96.47 kJ mol⁻¹). The energy absorbed may be measured directly by determining the absorption coefficient for the reaction system, although this procedure is somewhat difficult. The method is simplified greatly if it is possible to arrange matters in such a way that all the radiation is absorbed. A second method, more widely used, is to measure the amount of ionization produced by the radiation when it passes through the reaction vessel containing a gas such as air. The rate of production of ions is determined by applying to the system a sufficiently high electrical potential that the current reaches a limiting value (the *saturation current*). When this is done, the ions are swept out of the system as rapidly as they are produced by the radiation, and the rate of production of the ions can be calculated from the saturation current. To convert the rate of ion production into the energy absorbed per second, it is necessary to know what energy is dissipated in creating an ion pair; for air the value usually employed is 32.5 eV per ion pair for electrons, and 35 eV per ion pair for other types of radiation.

A third method for determining the rate of energy absorption, and the one most commonly used, involves allowing the radiation to pass into a reaction vessel that contains reactants, the reaction rate of which can be followed and related in a known manner to the radiation absorbed. Sometimes the oxidation of ferrous sulfate in 0.4 M sulfuric acid is used for this purpose.

In the study of overall reactions initiated by ionizing radiations, two quantities are quoted frequently and lead to important information about mechanisms. The first is the *ion-pair yield* or *ionic yield*, which is denoted as M/N : this is the ratio of the rate of production of product molecules to the rate of formation of ion pairs. The second is the *G value*, which is the number of product molecules produced per 100 eV input. Both these quantities give some idea of chain lengths.

9.4.1 Radiation-Chemical Primary Process

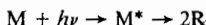
A process in which high-energy radiation interacts with a chemical substance with the formation of atoms and free radicals, as well as ions, is known as a *radiation-chemical primary process*. The species produced subsequently undergo secondary processes, which may involve chains. In a photochemical reaction, the primary process usually occurs in only two steps, the formation of an electronically excited species and its subsequent dissociation into atoms. In a radiation-chemical reaction, on the other hand, the primary step involves the formation of ions together with the formation of



The positive ion continues on its way with little deflection and ionizes other molecules. The ejected electron often has sufficient energy to bring about ionization of additional molecules, by the mechanisms described above.

Photons differ from the material particles in that they are absorbed and annihilated usually in a single elementary act, instead of undergoing a stepwise loss of energy. Three distinct processes are of importance in reactions brought about by photons:

1. Formation of an excited molecule, which may dissociate into atoms and radicals:



This is the type of process occurring in photochemical initiation.

2. Electromagnetic radiations having energy sufficient to cause ionization but not greater than about 0.5 meV (5×10^5 eV) lead to the ejection of an electron with the formation of the positive ion:



There is complete transfer of the energy of the radiation to the electron, which therefore is ejected with an energy equal to that of the photon minus the ionization energy.

3. At energies between about 10 keV (10^4 eV) and 1 meV (10^6 eV) there is Compton scattering of the photon with ejection of the electron and the formation of a positive ion:

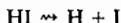


The difference between this process and that of mechanism 2 is that the photon is not annihilated but continues on its way with a different frequency. The scattered photon is liable to be absorbed by subsequent molecules in its path.

In spite of the fact that the details of radiolytic initiation processes are somewhat complex, the overall result is usually fairly simple. In hydrogen, for example, the initiation process with most types of radiation is largely the production of atoms,



Similarly, with hydrogen iodide the main process is



In all cases, however, the production of ions also must be taken into consideration.

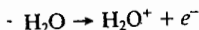
Reference has already been made, in Section 5.1.1 (see Fig. 5.1), to the technique of producing hydrogen atoms in an electric discharge. In such a discharge a stream of electrons is emitted by the cathode, and the mechanism of atom formation is therefore as previously discussed with reference to Fig. 9.9.

9.4.2 Pulse Radiolysis

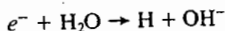
In radiation chemistry the analogue of flash photolysis is *pulse radiolysis*: instead of the light flash there is a pulse of high-energy radiation, usually a beam of electrons. For example, linear electron accelerators have been used to give pulses of 1–10 MeV with durations usually of a few microseconds; much shorter pulses, of the order of a nanosecond, also have been obtained. These pulses have been applied to a considerable number of substances, and the results have led to valuable information about the existence of transient intermediates and to the rate constants for the elementary reactions that occur.

9.4.3 Hydrated Electrons

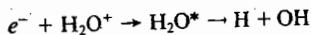
There has been much interest in the radiolysis of liquid water. With various types of ionizing radiation, the first step is the ejection of an electron:



If the electron produced were to remain free, it rapidly would undergo processes such as the following:



and



However, in 1953 it was suggested on theoretical grounds by Platzman³⁹ that an electron might become somewhat stabilized by being hydrated. He suggested that the hydrated electron, given the symbol e_{aq}^- , would have a blue color and that the enthalpy of hydration would be about 200 kJ mol^{-1} ($\approx 2 \text{ eV}$). Being stabilized in this way the hydrated electron would have a much longer life than a free electron and would survive long enough to react with solutes, bringing about their reduction.

Indirect support for the existence of the hydrated electron was provided by kinetic studies. For example, the reduction of certain substances in solution showed an ionic-strength effect (Section 6.3.4) that indicated the reducing species to have a negative charge;⁴⁰ it therefore could not be the H atom. Thus, it appeared that the reducing species are electrons, which must be hydrated since free electrons would react rapidly with water and would not survive long enough to undergo reaction with solutes. Another application of the ionic-strength method has been to the reaction between the hydrated electron and lysozyme.⁴¹ The reaction was found to be diffusion controlled, and the ionic-strength effect indicated that the hydrated electron was interacting with various ionized states of the lysozyme molecule; as the pH is increased the molecule is deprotonated.

Direct evidence for the existence of the hydrated electron was provided by pulse radiolysis studies of liquid water, in which the transient spectrum of the species was observed.⁴² The spectrum showed a maximum absorption at about 700 nm, and this is consistent with the blue color predicted by Platzman.

Since the 1960s many investigations have been made of the properties of the hydrated electron. Some rate constants for reactions involving it are listed in Table 9.2.

- 9.10. Confirm that the rate constant given in the previous problem for the reaction between ϵ_{aq}^- and $\text{Fe}(\text{CN})_6^{3-}$ is consistent with a diffusion-controlled reaction. Use the following data:

	Radius/nm	Diffusion coefficient/ $\text{cm}^2 \text{ s}^{-1}$
ϵ_{aq}^-	0.27	4.8×10^{-5}
$\text{Fe}(\text{CN})_6^{3-}$	0.28	9.0×10^{-6}

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Homogeneous Catalysis

One special type of catalysis, *heterogeneous catalysis*, has already been treated in Chapter 7. The present chapter is concerned with various types of catalysis that occur in a single phase and which therefore are referred to as *homogeneous catalysis*. Important examples are catalysis by acids, bases and enzymes.

The concept of catalysis was first recognized clearly by Berzelius¹ who, however, regarded the effect as due to a special type of force that the catalyst exerted on the reaction system. More understanding of catalysis was provided by a series of investigations carried out by Ostwald between 1888 and the turn of the century. In the course of his work, Ostwald proposed several useful definitions of a catalyst. One of them is that a catalyst is²

any substance that alters the velocity of a chemical reaction without modification of the energy factors of the reaction.

A later definition is that³

a catalyst is any substance that alters the velocity of a chemical reaction without appearing in the end product of the reaction.

Another definition, perhaps the best known, is that a catalyst is⁴

a substance that changes the velocity of a reaction without itself being changed by the process.

All these definitions are quite acceptable and are essentially equivalent to one another. They are consistent with the definition approved in 1981 by the International Union of Pure and Applied Chemistry:⁵